A. The two gen-ai used are Claude and Copilot.

B. Claude was accessed on the web, while Copilot was accessed on my laptop.

C. Write a C program that picks a secret number between 1 and 10. The user gets up to 3 tries to guess the number. Requirements: Use a fixed secret number (for example, 7). Use printf to prompt the user and scanf to read their guess. Use a while or for loop to let the user guess up to 3 times. Use if-else statements to check if the guess is too high, too low, or correct. Stop the loop immediately if the user guesses the correct number. Print a final win or lose message.

   Example Output

   Guess a number between 1 and 10.

   Attempt 1/3. Enter your guess: 5

   Too low! Try again.

   Attempt 2/3. Enter your guess: 8

   Too high! Try again.

   Attempt 3/3. Enter your guess: 7

   Correct! You win!

D. Copilot:

```c
#include <stdio.h>

int main() {
    int secret = 7;
    int guess;
    int attempts = 3;

    printf("Guess a number between 1 and 10.\n");

    for (int i = 1; i <= attempts; i++) {
        printf("Attempt %d/%3d. Enter your guess: ", i, attempts);
        scanf("%d", &guess);

        if (guess == secret) {
            printf("Correct! You win!\n");
            return 0;
        }
        else if (guess < secret) {
            printf("Too low! Try again.\n");
        }
        else {
            printf("Too high! Try again.\n");
        }
    }

    printf("Sorry, you lose! The correct number was %d.\n", secret);

    return 0;
}
```

e. Copilots code is missing error checking and could throw an error if an int isn't entered. Copilots code has faster execution time as it uses less control structures and variables. Copilots code has better space complexity as it uses less lines of code to accomplish the same task. Although copilots code is overall better, it is harder to read and understand, giving it worse maintainability than Claude.

Claude:

```c
#include <stdio.h>

int main() {
    int secret = 7;
    int guess;
    int attempts = 3;
    int won = 0;

    printf("Guess a number between 1 and 10.\n");

    for (int i = 1; i <= attempts; i++) {
        printf("Attempt %d/%d. Enter your guess: ", i, attempts);
        scanf("%d", &guess);

        if (guess == secret) {
            printf("Correct! You win!\n");
            won = 1;
            break;
        }
        else if (guess < secret) {
            printf("Too low! Try again.\n");
        }
        else {
            printf("Too high! Try again.\n");
        }
    }

    if (!won) {
        printf("Out of tries! The number was %d. You lose.\n", secret);
    }

    return 0;
}
```

e. Claude's code is missing error checking and could throw an error if an int is not entered. Claude's code has slightly more execution time than copilots as it includes an extra if statement and variable. Claudes code has worse space complexity as it uses more space than Copilot. Claude's code is more maintainable than copilots as the logic is easier to read and understand.

f. I picked Copilots code as it accomplishes the same task with less lines of code. It also has better execution time and space complexity.

g. I added comments throughout Copilots code, making it more readable. I could not find a way to improve the execution time. I could not find a way to improve space complexity either without removing some functionality of the program.